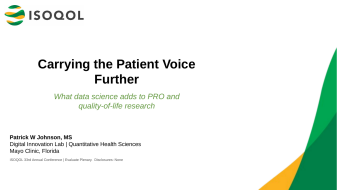
SLIDE 1

No speaker notes



SLIDE 2

Last year, AI was everywhere at this meeting. It was the conference theme, and a lot of the conversation was about understanding these tools - what machine learning and large language models can do, how we train and evaluate them, how we make sense of their predictions, and where they might fit in quality-of-life research.

A year later, I think the questions are a little different.

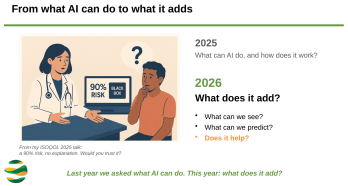
It's not just, **what can AI do?**

It's **what does it add?**

What new information can we use? What can we predict? And when we put it into practice, does it actually help?

And I think that's a useful way to look at methods in general. Not *is this new?* or *is this sophisticated?* but **what does it give us that we didn't have before?**

So before I get into the examples, I should probably be clear about what I'm calling data science."



SLIDE 3

This was probably the hardest slide for me to put together, because if you ask ten people what data science means, you'll probably get ten different answers.

I tend to think of it as bringing together a few different traditions.

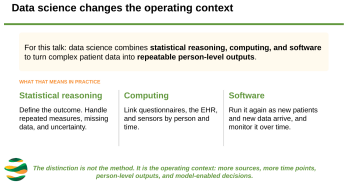
Some of it is statistical reasoning. We still need to define what we're estimating, deal with missing data and repeated observations, and understand the uncertainty.

Some of it is simply that computing changed. We have much more data, faster hardware, GPUs, cloud computing. Things that weren't practical computationally aren't anymore.

And some of it is software engineering. Not just getting the code to run once, but making it reproducible, testable, maintainable, and able to run again when new data show up.

What becomes interesting is when a project needs all three. Maybe we're linking several kinds of patient data, updating something over time, and producing an output that has to work reliably for the next patient, not just the patients in the original dataset.

That's generally the context in which I'm using the term data science today.



SLIDE 4

And once you describe data science that way, there's obviously a lot of overlap with biostatistics.

I don't think there's a clean line between the two. I think of these more as different centers of gravity.

Across the top is the lifecycle of a project, from the scientific question through the data, modeling and validation, and eventually implementation and evaluation.

Someone trained in biostatistics may naturally put more weight on study design, estimands, inference, and uncertainty. Someone coming from computer science or software engineering may put more weight on computational scale, algorithms, software systems, or what it takes to keep something running after the initial analysis.

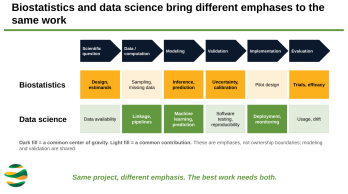
But none of those are exclusive. Biostatisticians build software and link data. Data scientists think about uncertainty. Everybody models and validates.

And in my own work, some of the projects I'm happiest with have been collaborations with biostatisticians. We may come into the same problem asking slightly different questions, and that's usually useful. One person may be pushing on whether we've defined the scientific question correctly, while somebody else is thinking about how to make the data or the system actually work.

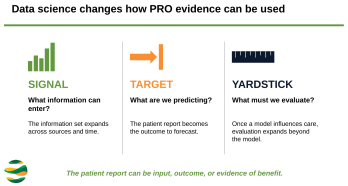
A lot of these problems benefit from having both perspectives in the room early, rather than handing the project from one discipline to another at the end.

There's a lot of shared territory here. Where the emphasis falls depends on the problem, the team, and frankly, who you ask.

What I care about more is what becomes possible when we bring these pieces together.



SLIDE 5



For patient-reported data, I think about this in three ways.

First, **signal**. The patient report is information going into the analysis, alongside other data and potentially over time.

Second, **target**. The patient report can itself be what we're trying to predict.

And third, **yardstick**. Once a model starts influencing care, predicting well isn't enough. We also need to know whether anything got better for the patient.

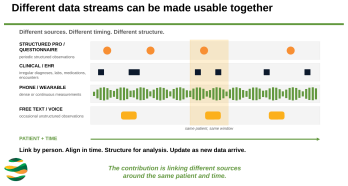
The patient report can be the input, the outcome, or the evidence that what we did actually helped.

SLIDE 6



I'll start with the PRO as signal, or simply the information we have available to work with.

SLIDE 7



What has changed more recently is the range of data we can bring into the same analysis.

A PRO might be collected every few weeks. The EHR gives us labs, medications, diagnoses, and encounters. Phones and wearables can generate measurements throughout the day. And now we increasingly have access to things like imaging, genomics and other omics data, free text, and voice.

Some of those data have always existed in one form or another. What's different is our ability to store them at scale, extract useful information from them, and actually use them together.

And these aren't just more variables. They're fundamentally different kinds of data. An image looks nothing like a questionnaire. A continuous sensor stream looks nothing like a lab result. Genomic data have a completely different structure again.

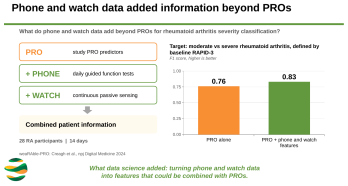
So part of the work is making those sources usable together: linking them to the same patient, aligning them when timing matters, and finding useful representations of much more complex data.

That opens up a much richer view of the patient. But more data doesn't automatically mean better information. A wearable giving me thousands of measurements a day isn't necessarily telling me something more important than a questionnaire collected once a week.

The scientific question still has to determine what information is useful.

Longitudinal analysis itself isn't new. What's changed is the range, scale, and complexity of information we can now bring together around the patient.

SLIDE 8



Here's a concrete example of what I mean.

weaRAble-PRO was a 14-day observational study in rheumatoid arthritis. The investigators were interested in whether data collected remotely during daily life could add information beyond patient-reported measures

when characterizing rheumatoid arthritis and its severity.

There were really three different types of information being collected.

Patients completed PRO measures during the study. They also used an iPhone each day for a set of guided functional tests. These included walking, wrist range of motion, a digital dexterity test, and getting up from a chair or from lying down. The phone's sensors and touchscreen turned those tasks into quantitative measures.

The Apple Watch was doing something different. It collected data passively as people went about their daily lives, which could then be used to characterize things like physical activity and sleep.

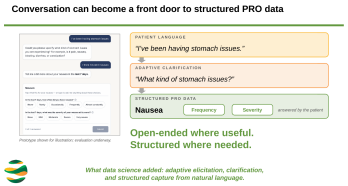
The analysis shown here focused on the 28 participants with rheumatoid arthritis and asked whether those sources could help distinguish moderate from severe disease, as defined by their baseline RAPID-3.

With the PRO measures alone, the F1 score was .76. Adding the passive watch data increased that to .79, and combining the PROs with both the phone and watch features gave the best performance, at .83.

What I like about this example is that these are three quite different ways of learning about the same patient: what the patient reports, how they perform on a guided task, and what happens during everyday life.

Data science gives us a way to bring those different sources into the same analysis.

SLIDE 9



There's another way the signal can change. Instead of adding another data source, we can change how the patient tells us what they're experiencing.

This is a prototype our team has been developing for symptom assessment. The idea is to let the patient start in their own language rather than requiring them to start with a predefined item.

A patient might say, "I've been having stomach issues." That's not yet very useful as structured data, so the system asks what they mean. The patient says nausea. Once the symptom is identified, we can move into the structured questions we need, like frequency and severity, and those answers still come directly from the patient.

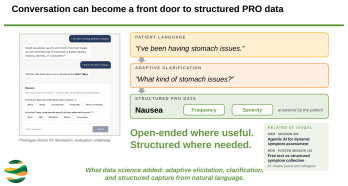
The goal isn't to replace structured measurement with a chatbot. It's to use conversation where it helps, while preserving structure where we need it for measurement and analysis.

And there are some interesting measurement questions underneath this. Does asking about a symptom conversationally change how someone responds? How much can the system clarify before we've changed what we're trying to measure? And if the system interprets something the patient said, how do we make sure the patient can see it and correct it?

Those aren't really chatbot questions. They're measurement questions. And they're another reason I think this work needs people who understand PROs involved from the beginning.

Open-ended where useful, structured where needed.

SLIDE 10



If you're interested in this, Dr. Amylou Dueck is presenting this work in Session 303 on Wednesday, and there's a related poster from Monday on free text versus structured symptom collection.

Both go much deeper into the questions I'm only touching on here.

SLIDE 11

The PRO as target

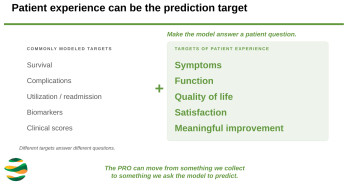
What are we predicting?

"So far, the patient report has been part of the information entering the system.

Now let's put it on the other side.

What are we predicting?"

SLIDE 12



The patient report can also move to the other side of the model and become what we're trying to predict.

A lot of predictive work in healthcare naturally starts with outcomes that are already available in clinical data: survival, complications, readmissions, biomarkers, clinical scores. And those are obviously important.

But there's also a practical reason those outcomes show up so often. They're there. Mortality is in the record. A readmission is in the record. A lab value is already structured and relatively straightforward to model.

Whether someone can get back to the activities that matter to them, whether they're sleeping better, whether their symptoms are tolerable, or whether they feel treatment was worth it may not be.

So part of choosing a target is deciding what question we actually want the model to answer.

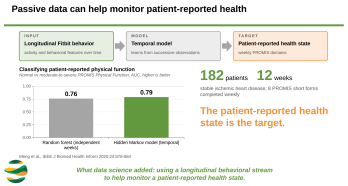
If the question is about how the patient will feel or function, then symptoms, quality of life, satisfaction, or meaningful improvement can become the target instead.

And this is where collaboration with quality-of-life researchers really matters. If we simply start with whatever outcome is easiest to pull from the data, we can end up building very sophisticated models for outcomes that aren't necessarily the ones patients care most about.

People in this room bring expertise in deciding which aspects of patient experience matter, how to measure them, and what a change in them means. That expertise needs to come in when we're defining the problem, not after the model has already been built.

The model can tell us how well we predicted the outcome we gave it. It can't tell us whether we chose the right outcome in the first place.

SLIDE 13



These investigators asked whether passive wearable data could help track what patients themselves were reporting about their health.

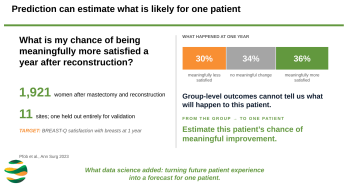
They followed 182 people with stable ischemic heart disease for up to 12 weeks. Patients wore a Fitbit continuously and completed eight PROMIS measures each week.

The Fitbit captured activity and physiologic data, while the patient reports defined the health states the models were trying to recognize.

The interesting possibility is what you could eventually do with that. PROs take effort to collect, and the people who stop responding may be exactly the people we're most worried about. Passive data could potentially help us monitor between assessments, identify when another report would be useful, or flag someone who may be falling through the cracks.

The patient report still defines what we're trying to understand. The model gives us another way to follow it.

SLIDE 14



This takes the target idea one step further. Instead of monitoring a patient's current state, the investigators asked whether they could predict a future patient-reported outcome for an individual patient.

They used data from 1,921 women undergoing mastectomy and breast reconstruction. Before surgery, they had clinical characteristics, patient characteristics, and baseline PROs. The outcome was whether satisfaction with

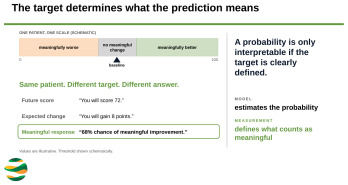
breasts would meaningfully improve or worsen one year later.

At the population level, some women improved, some worsened, and some stayed about the same. But the prediction is much more personal: given what we know about this woman before surgery, what is her chance of meaningful improvement or worsening?

And the models did reasonably well, with AUCs generally in the .8s.

The PRO here isn't just something collected later to describe the outcome. It's the future patient experience the model is trying to predict.

SLIDE 15



There's another place where collaboration with quality-of-life researchers becomes important.

Suppose I tell a patient she has a 68 percent chance of meaningful improvement. That sounds straightforward, but somebody had to define what meaningful improvement actually means.

We could predict her score at one year. We could predict how much she will change. Or we could predict whether that change crosses a threshold we consider meaningful.

Those are different targets, and choosing among them isn't primarily a modeling decision. It depends on how the outcome is measured and how we interpret change in that measure.

This is where modelers need people who understand the measurement. The model can estimate the probability, but it needs a well-defined target first.

And if that prediction is eventually going to be used with a patient, getting that definition right matters quite a bit.

SLIDE 16

The PRO as yardstick

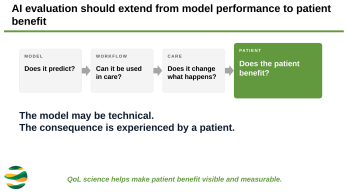
What must we evaluate now?



And once a prediction actually starts influencing care, we have a different question to answer.

Did it help the patient?

SLIDE 17



Once a model starts influencing care, technical performance is only the beginning of the evaluation.

It has to work well enough to trust. But then someone has to actually use it. It has to fit into a clinical workflow, influence a decision or an action, and eventually affect something we care about.

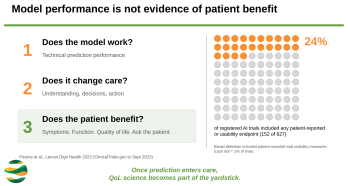
Those are different stages, and success at one doesn't guarantee success at the next. A well-performing model may never get used. A model that gets used may not change care. And changing care doesn't necessarily mean the patient is better off.

Once we get to that last question, the outcomes also change. We may care about symptoms, function, quality of life, or the experience of making a difficult decision.

That's where the quality-of-life community has a very specific role. We know how to define those outcomes, how to measure them, and how to determine whether a difference is meaningful.

The model may be technical, but the consequences of using it are experienced by the patient.

SLIDE 18



And these really are different kinds of evidence.

An AUC can tell us whether a model discriminates. Calibration can tell us whether the probabilities are believable. Neither tells us whether clinicians used the prediction, whether it changed a decision, or whether the patient was better off because of it.

If we care about symptoms, function, quality of life, or the experience of care, at some point we have to measure those things.

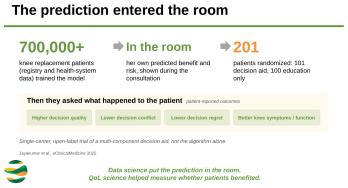
And that still isn't routine. In this review of 627 registered interventional AI trials, only 24 percent included any patient-reported or usability endpoint.

That's a broad definition. It includes more than validated PROMs, so I wouldn't interpret the other 76 percent as studies that ignored patients or only measured technical performance.

But even with that broad definition, patient-reported evidence was included in only about a quarter of these trials.

For a field increasingly talking about patient-centered AI, I think that's an important gap, and one this community is particularly well equipped to help fill.

SLIDE 19



Here's an example where that evaluation was carried further.

The underlying models were trained on data from more than 700,000 people who had undergone knee replacement. They combined clinical information and PROs to estimate an individual patient's likelihood of benefit and risk from surgery.

Those predictions were then built into a decision aid, alongside patient education and an assessment of the patient's preferences, and used during the consultation.

The investigators randomized 201 patients to receive either the full decision aid or education alone.

And they measured quite a bit more than whether the prediction was accurate. They asked about decision quality and decision conflict. They followed patients for decision regret. And they measured knee symptoms and function.

Patients who received the decision aid had higher decision quality, less decision conflict, less decision regret later on, and better knee symptoms and function at follow-up.

I like the range of outcomes here. Benefit isn't defined only as a better knee score. It's also whether the patient felt informed and comfortable with the decision they made, and whether they regretted that decision later.

The prediction entered the clinical encounter, and the evaluation followed through to the patient's experience and health.

SLIDE 20



Let me come back to the three roles we started with.

As signal, the information available to us expanded. We could combine patient reports with clinical data, phones, wearables, and other sources.

As target, the patient report became something we could estimate or forecast, whether that's monitoring someone's current health or predicting how they might feel after treatment.

And as yardstick, once those predictions entered care, the evaluation expanded too. We had to ask not only whether the model worked, but whether the patient benefited.

Across all three, the patient report supplied the signal, defined the target, or helped us judge the benefit.

And none of this makes the other disciplines in this room less important. If anything, using patient-reported data in these new ways makes good measurement, qualitative research, and careful statistical thinking even more important.

SLIDE 21

Data science can carry the patient's report further than it has ever gone.

But it must stay anchored to the patient's report.



Data science gives us new ways to carry the patient voice.

We can connect it with new kinds of data. We can make patient experience something we try to predict. And when those predictions enter care, we can use patient-reported outcomes to ask whether they actually helped.

That can take the patient's report further than it has gone before.

But as we do that, it still has to stay anchored to what the patient actually tells us.

We can expand it, model it, and build predictions around it. We shouldn't mistake any of those things for the patient's report itself.

Thank you.